

Narrative / Symbolic Space Analyzer

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What is non-geocodable (symbolic) space?

GIS_main maps geocodable space — real places that resolve to map coordinates (New York; Upson County, GA). But most of what happens in a narrative unfolds in non-geocodable space: kinds of place that carry social / symbolic meaning but have no lat/long — the house, the field, the threshold of a door, the woods. “Kitchen vs. field” is not a coordinate; it is a KIND of space. This tool takes the failed geocode and treats it as the SIGNAL, not the noise.

The space types

Each place word is classified into a space TYPE via an editable gazetteer, lib/symbolic_space_typology.csv (WordNet-seeded): domestic/interior, field/labour, wild/forest, threshold/liminal, royal/court, sacred, market/public, water/passage. Edit it to add your domain’s words (e.g. jail, courthouse, porch).

Two analyses — one input file

DYNAMIC — movement. How social actors move between kinds of space over the narrative (door → house → field → woods) as a directed movement graph. The movement is the finding.

STATIC — distribution. A social attribute (gender, race, class) cross-tabbed against the kind of space → a heatmap with chi-square, Cramér’s V and residuals (women ↔ kitchen; Black ↔ wild).

What it expects in INPUT — one CSV

Both analyses read the SAME file, different columns: Location (both), Sequence (DYNAMIC), Attribute (STATIC). Example — the lynching narrative as a table:

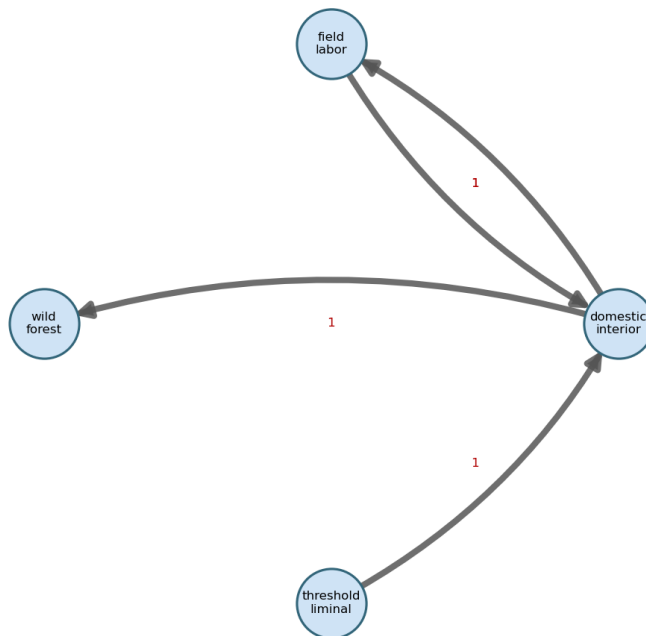
step	actor	race	location
1	accused	Black	door
2	woman	white	door
3	accused	Black	house
4	accused	Black	field
5	mob	white	field
6	accused	Black	house
7	accused	Black	woods
8	accused	Black	tree

Map: Location → location, Attribute → race, Sequence → step. Tick both boxes, RUN → the movement graph AND the heatmap from this one file.

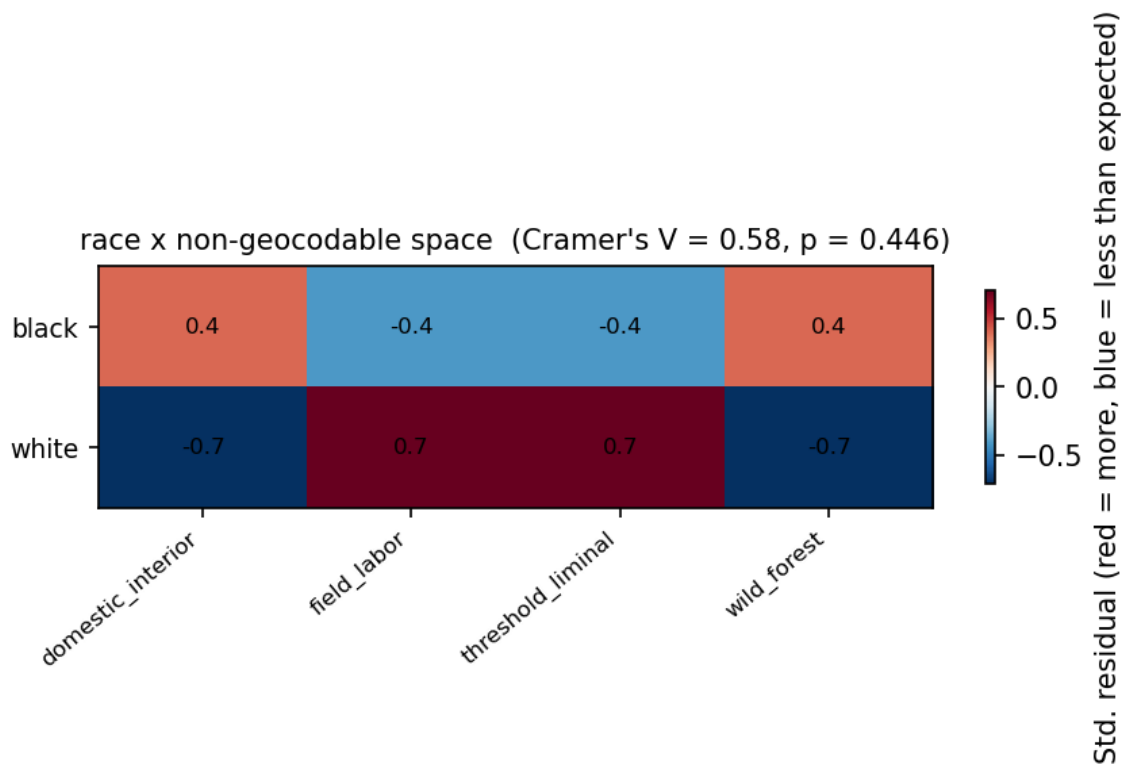
What it produces in OUTPUT (on this example)

DYNAMIC — the movement graph. The accused crosses the domestic threshold, flees to the field, is returned to the domestic for identification, and is executed in the wild:

Movement through non-geocodable space (the lynching narrative)



STATIC — the race × space heatmap (standardized residuals: red = more than chance, blue = less):



Files written: symbolic_space_transitions.csv + symbolic_space_movement.png (dynamic);
symbolic_space_<attribute>_x_space.csv (+ _residuals.csv) + _heatmap.png (static).

Where the CSV comes from

1. Bring your own data. Any CSV with a location column (+ attribute and/or order) works — no text processing. This is how the lynching project works.
2. From a text corpus. The SVO extractor (SVO_main: who-did-what-where) gives actor + location + order; add the actor's gender with the gender annotator. The “GUIs available for more analyses” menu opens these tools.

Reading the results

Movement graph: thick arrows are the canonical moves; across many texts the recurrent path is the genre's spatial grammar. Heatmap (residuals): red = that attribute in that kind of space MORE than chance, blue = less; Cramér's V near 1 = a strong who-is-where association.

Further readings — more TIPS files

TIPS_NLP_GIS_Geocoding.pdf · TIPS_NLP_csv files - Problems & solutions.pdf